

Programming Exercise: Web Scraping & Data Analysis

Foundation of Data Science
Shiraz University

Overview

This assignment consists of two main parts: (1) scraping a data table from Wikipedia, and (2) performing data analysis, and visualization on the scraped data. You will work with real-world population data to practice essential data science skills.

1 Web Scraping

Scrape a data table from a Wikipedia article that contains at least 50 rows and extract all columns from the table. For this assignment, use the following Wikipedia page:

[https://en.wikipedia.org/wiki/List_of_countries_by_population_\(United_Nations\)](https://en.wikipedia.org/wiki/List_of_countries_by_population_(United_Nations))

This page contains a well-structured HTML table listing countries by population, with over 200 entries.

1.1 Requirements

Your scraping code must:

1. Scrape the data table from the Wikipedia article
2. Ensure the table contains at least 50 rows
3. Extract all columns present in the table
4. Implement error handling using `try--except` blocks
5. Store the scraped data as a list of dictionaries
6. Convert the data into a `pandas.DataFrame`
7. Save the raw dataset as `scraped_data_raw.csv`

1.2 Technical Approach

- Use the `requests` library to fetch the HTML content
- Parse the HTML using `BeautifulSoup`
- Identify the main data table by verifying row count ≥ 50
- Handle potential errors gracefully
- Document your code with clear comments

1.3 Required Libraries

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
```

2 Data Analysis

Perform the following analyses on your cleaned dataset:

2.1 Descriptive Statistics

1. Calculate summary statistics for population data (mean, median, std, min, max)
2. Identify the top 10 most populous countries
3. Identify the bottom 10 least populous countries
4. Calculate the total world population from the dataset

2.2 Comparative Analysis

1. If the table contains multiple years, calculate:
 - Population growth rates for each country
 - Countries with highest and lowest growth rates
 - Average global population growth rate
2. Analyze population distribution:
 - What percentage of countries account for 50% of world population?
 - Create population categories (e.g., small: < 10M, medium: 10M-100M, large: > 100M)
 - Count countries in each category

2.3 Data Visualization

Create the following visualizations:

1. **Bar Chart:** Top 15 countries by population with appropriate labels and title
2. **Histogram:** Distribution of population across all countries (use logarithmic scale if needed)
3. **Pie Chart** : Population share of top 10 countries vs. rest of the world